

Education

New York University, New York, NY

Sep 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

Experience

- CampusMesh** Jan 2026 – Present
Software Engineer
New York, US
 - Owned delivery of scalable **Node.js microservices** for member-facing academic workflows across study groups, messaging, calls, notifications, and profile management, backed by **MySQL (Sequelize)** and **DynamoDB**.
 - Deployed **highly-available serverless services** on AWS using **Serverless Framework** with **Lambda**, **API Gateway**, **S3**, and **CloudFront** across **4 environments**; enforced security with **KMS-managed keys**, **SSM Parameter Store**, and **least-privilege IAM**.
 - Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.
- Amazon** May 2025 – Aug 2025
Software Development Engineer Intern
Seattle, US
 - Owned end-to-end delivery of a secure UI console for fulfillment center teams; presented **system design** for approval and established Role Based Access (**4+ roles**) using **OAuth**, **Midway SSO**, and **HELIS**.
 - Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, DynamoDB, S3, and IAM, reducing provisioning time by **70%** and enabling **multi-stage, multi-region CI/CD** with **Git**-based build automation.
 - Built a **React 18** frontend with optimized API integration, advanced filtering, and client-side caching, reducing page load time by **40%**.
 - Engineered maintainable backend services in **Java** using **Coral**, **Smithy**, **Dagger**, and **CBOR**; applied **OOP design patterns**, reduced execution time by **35%**, and wrote **unit and integration tests** with **JUnit** and **Mockito**.
- New York University (Novel AI Technologies, Inc.)** Aug 2024 – Present
Software Developer (On-Campus)
New York, US
 - Architected an **NSF-funded (\$100K)** AI-powered health app for **iOS** and **Android**, processing real-time wearable data; deployed cloud-hosted AI models for health metrics analysis and food image processing.
 - Engineered ETL pipelines transforming **NoSQL** into **PostgreSQL** using **concurrency control and locking mechanisms**; processed **250K+** records and powered **AWS QuickSight** dashboards for real-time health analytics.
 - Developed a financial analytics platform (**React**, **Node.js**, **REST APIs**) with **Stripe billing**, **RBAC**, and row-level security; deployed **Docker** containers on **EC2**, generating **\$50K+** revenue across regulated client workflows.
 - Streamlined underwriting via broker portal integrated with insurer platform; implemented **document and media processing** and **WebSocket-based real-time communication**.
- PricewaterhouseCoopers (PwC)** Aug 2021 – Aug 2024
Senior Software Engineer
Bengaluru, India
 - Developed a **distributed backend** on **Microsoft Azure** for a health platform, integrating with **Microsoft CRM** and implementing event-driven processing with fault-tolerant retries over **SQL Server**.
 - Deployed an **NLP based OCR neural model** on Azure to automate structured data extraction, reducing data processing time by **75%**.
 - Optimized data ingestion with **parallel processing** for **100K-row** Excel feeds; reduced execution time from **5 hours to 1 hour**.

Projects

- Advanced Expense Tracker (Java Desktop Application)** | *Java, JavaFX, JDBC, SQLite, JUnit, JFreeChart*
 - Architected a modular **Java application** supporting individual and group expenses with secure login, applying **OOP principles** and **MVC & DAO design patterns**.
 - Implemented concurrent transaction handling using **multithreading** and synchronized blocks for thread-safe updates; added **JUnit** tests for core business logic.
- Hospital Management System (Full-Stack Application)** | *React, Node.js, Express, MongoDB, Jest*
 - Built a **full-stack healthcare platform** using **React**, **Node.js**, **Express**, and **MongoDB** to manage appointments, patient records, lab reports, and medical history with **role-based access control**.
 - Applied **relational and document database** patterns across MongoDB collections and normalized schemas; wrote **API unit tests** for critical service paths.
- RoBERTa Fine-Tuning with Custom LoRA** | *Python, PyTorch, Hugging Face, AGNews*
 - Architected a **parameter-efficient fine-tuning (PEFT)** pipeline using custom **LoRA** to optimize a **RoBERTa-base** model for news classification, achieving **83.45%** accuracy on the private leaderboard.

Skills

- Programming Languages:** Java, C++, Python, TypeScript, JavaScript (ES6), SQL
- Web Technologies:** HTML, CSS, React, REST API, GraphQL API, Microservices, Node.js, Express.js, Sequelize
- Database:** RDBMS (MySQL, PostgreSQL, SQL Server), NoSQL (DynamoDB, MongoDB, Firebase), Dataverse
- Cloud:** AWS (CDK, EC2, S3, DynamoDB, API Gateway, Lambda, Cognito, IAM, SageMaker), Azure (Function Apps, OpenAI services, Storage)
- Tools:** Git, JUnit, Mockito, Docker, Kubernetes, CI/CD, Visual Studio, Postman, Power BI
- AI/ML:** LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning

Certifications

AWS Cloud Practitioner

Azure Data Fundamentals

Stanford Machine Learning (Coursera)